Course: AWS Developer Certification + Cloud Practitioner — 12 weeks (12 × 4hr sessions + labs)

Course Teacher: **Saim Safder** (Solutions Architect @ https://www.cloudidr.com/)

This course introduces students to the key principles of DevOps, a set of practices that combine software development (Dev) and IT operations (Ops) to improve collaboration, automate workflows, and enable continuous delivery. Students will learn how to bridge the gap between development and operations, focusing on automation, monitoring, infrastructure as code (IaC), containerization, and cloud services.

Target audience: Students, professionals, AWS Cloud Practitioners preparing for AWS Cloud Practitioner (foundational) and AWS Certified Developer – Associate (associate).
Format: Weekly lecture + hands-on lab (cloud console / IaC) + weekly quiz. Final week = project presentation + proctored-style final exam.
Prerequisites: Basic Linux, Git, programming (Python/Node), and the DevOps fundamentals from your original syllabus.

Fast references (exam format / official resources)

- **AWS Certified Cloud Practitioner (Foundational)** — 65 questions, **90 minutes**, multiple choice/multiple response. (official).
- **AWS Certified Developer — Associate** — 65 questions, **130 minutes**, multiple choice/multiple response; developer topics include Lambda, API Gateway, DynamoDB, SDKs, CI/CD, and deployment patterns.
- Use the official AWS exam guides and AWS Skill Builder learning plans for official domain weights and practice materials.

Exam alignment matrix (short)

- **Cloud Practitioner** (Foundational): Understand cloud concepts, AWS core services (EC2, S3, RDS, VPC), billing/pricing, security basics, AWS shared responsibility. (use official exam guide).

- **Developer – Associate**: Application development/deployment on AWS: Lambda, API Gateway, DynamoDB, S3, IAM roles/policies, CI/CD, CloudFormation/CDK, X-Ray/CloudWatch, container basics & ECR, SDK usage. Labs Weeks 3–9 map strongly to these domains.

Week-by-week syllabus (detailed)

Week 1 — Cloud foundations + AWS account, IAM basics

Learning goals: Cloud principles, AWS global infra, accounts & billing, IAM concepts (users/roles/policies). Map to Cloud Practitioner domains (cloud concepts, billing) and Dev Associate basics (IAM best practices).

Lecture topics: Cloud models, shared responsibility, AWS regions/availability zones, AWS pricing overview, IAM users/roles/policies, and least privilege.

Lab (deliverable): Create an AWS Free Tier account or classroom account; create IAM groups, users, and roles; enable MFA; implement an IAM role for EC2. (Submit: IAM policy JSON + screenshots + short justification document)

Assessment: Quiz (10 Q), practical rubric for IAM config.

Week 2 — Compute & networking on AWS (EC2, VPC basics, Load Balancer)

Learning goals: EC2 instance types, EBS, AMIs, security groups, VPC basics. Relevant to Cloud Practitioner (core services) and Developer (deploy/test apps).
Lecture topics: EC2 lifecycle, AMI creation, Elastic IPs, VPC subnets, route tables, NACL vs SG, ALB basics.
Lab (deliverable): Launch an EC2 instance (Cloud9 optional), create a custom AMI, configure a VPC with public/private subnets and an ALB serving a simple app. (Submit infra diagram + Terraform / CloudFormation minimal template + verification steps)
Exam mapping: Cloud Practitioner core services; Developer: environment setup for app deployment.

Week 3 — Object storage, databases & caching (S3, RDS, DynamoDB, ElastiCache)

Learning goals: Storage options, lifecycle policies, DynamoDB basics (NoSQL), connection patterns for apps. Developer exam emphasizes DynamoDB usage and SDK interactions.
Lecture topics: S3 features, presigned URLs, encryption, RDS vs DynamoDB, capacity modes, caching strategies.
Lab (deliverable): Build a small app that stores files to S3 (presigned PUT/GET) and stores metadata in DynamoDB. Provide code snippets using AWS SDK (Python or Node). (Submit

code repo + run instructions + short test cases)
Assessment: Quiz + code review.

Week 4 — Serverless fundamentals (AWS Lambda, API Gateway, event sources)

Learning goals: Lambda function lifecycle, triggers (S3, SQS, EventBridge), API Gateway REST & HTTP APIs. This is heavy Developer exam content.
Lecture topics: Lambda best practices (cold start, packaging, environment vars), API Gateway mapping, authorizers, error handling.
Lab (deliverable): Implement a serverless API: Lambda behind API Gateway, store results in DynamoDB, test synchronous and asynchronous invocation. Deploy with SAM/Serverless Framework/CDK. (Submit deployment template + test logs + CI snippet)
Assessment: Lab rubric covers correctness, security (IAM roles), observability (CloudWatch logs).

Week 5 — CI/CD and developer tooling (CodeCommit / CodeBuild / CodePipeline / CodeDeploy; GitHub Actions alternative)

Learning goals: Build automated pipelines for tests, builds, and deployments; understand deployment strategies (blue/green, canary). Developer exam includes CI/CD knowledge.
Lecture topics: Pipeline stages, buildspec, artifact stores (S3, ECR), container registry (ECR).
Lab (deliverable): Create a pipeline to build and deploy the serverless API from Week 4; include unit tests and automated deployment to a dev stage. (Submit pipeline YAMLs/Buildspec + evidence of automated deployment and rollback test)
Assessment: Pipeline run log + checklist.

Week 6 — Infrastructure as Code (CloudFormation & CDK)

Learning goals: Declarative vs imperative IaC; author CloudFormation templates and a CDK app (Python/TypeScript). Developer exam expects ability to use CloudFormation/CDK for deployments.

Lecture topics: CloudFormation templates, change sets, CDK constructs, stacks vs nested stacks, drift detection.

Lab (deliverable): Convert the Week 2 or 4 infra into CloudFormation or CDK (preferred). Include parameterization and outputs. (Submit template/CDK repo + deployment commands + verification)

Assessment: IaC review rubric (idempotence, modularity, least privileges for created resources).

Week 7 — Observability & debugging (CloudWatch, X-Ray, CloudTrail, logging)

Learning goals: Implement logging, metrics, distributed tracing; debug serverless and containerized apps. Developer exam tests CloudWatch and X-Ray knowledge

Lecture topics: CloudWatch metrics/alarms/dashboards, logs insights, X-Ray traces, structured logging, CloudTrail for auditing.

Lab (deliverable): Add tracing to Lambda functions, create dashboards and alarms for key metrics, simulate a failure and produce an incident postmortem. (Submit dashboards + sample traces + incident report)

Assessment: Incident report + dashboards.

Week 8 — Containers, registries, EKS/ECS, ECR & deployment patterns

Learning goals: Build, push images to ECR; deploy to ECS Fargate or EKS (intro); container networking and secrets. Developer exam covers container deployment basics.
Lecture topics: ECR lifecycle, task definitions, Fargate vs EC2 launch types, CICD for containers.
Lab (deliverable): Containerize the app, push to ECR, deploy to ECS Fargate service with ALB. Optionally show a basic EKS deployment (minikube/eksctl). (Submit Dockerfile + task def + verification endpoints)
Assessment: Container production readiness checklist.

Week 9 — Security in development pipelines & app auth (Cognito, KMS, Secrets Manager)

Learning goals: Secure CI/CD, secrets management, authentication/authorization patterns for apps (Cognito, IAM roles). Developer exam includes identity, encryption, secrets.
Lecture topics: KMS key management, envelope encryption, Secrets Manager rotation, Cognito user pools & identity pools, OAuth flows.
Lab (deliverable): Integrate Cognito auth into the API, store secret config in Secrets Manager, implement KMS-encrypted data at rest. (Submit auth flow diagram + config + test users)
Assessment: Security checklist and short viva.

Week 10 — Machine Learning & AI services (SageMaker, Rekognition, Lex, Comprehend) (requested focus)

Learning goals: High-level use of AWS ML/AI services to add features to apps (recommendations, image analysis, chatbots). This week supports the Cloud Practitioner's AI/ML awareness and Developer tasks for integrating ML APIs.
Lecture topics: SageMaker concepts (notebooks, training, endpoints), Amazon Rekognition

(image analysis), Amazon Lex (chatbots), Comprehend (NLP), how to call these via SDK and secure endpoints.
Lab (deliverable): Build a small proof-of-concept: (a) Deploy a SageMaker endpoint (or use built-in inference) to classify a dataset OR use Rekognition to analyze uploaded images and store results; (b) build a Lex basic chatbot integrated with the API from Weeks 4–5. (Submit architecture diagram + code snippets + demo recording)
Assessment: POC demo + reflection on production readiness and cost implications.

Week 11 — Professional Development & Soft Skills (CVs, interviews, exam preparation)

Learning goals: CV tailoring for cloud roles, interview mock sessions, exam strategies, exam scheduling, study plan for Cloud Practitioner and Developer Associate. Also review exam domains and common pitfalls.

Activities: CV clinic (students bring CVs), 1:1 mock technical interviews, tips for multiple-choice exam strategy, exam scheduling (Pearson VUE). Provide official study resources and practice exams.

Week 12 — Project presentations & Final Exam (course exam)

Project presentation: Each team presents capstone (see next section for full project spec). Evaluation: architecture, code, CI/CD, security, monitoring, cost optimization, demo.
Final exam: Proctored 2-part exam option:

- **Cloud Practitioner (Foundational)**: 65 Q, 90 minutes — a pass/score report used for certification readiness. (use official practice set as warmup).
- **Developer (Associate)**: 65 Q, 130 minutes — instructor-provided mock exam based on AWS blueprint and official guide.

Capstone Project (detailed) — mandatory for Developer exam pathway

Project title (example): “Serverless Photo-Tagging & ChatOps App”
Objective: Build a production-grade application that demonstrates developer skills across AWS: serverless API, storage, NoSQL DB, user auth, CI/CD, observability, optional ML feature.
Minimum requirements (deliverables):

1. **Application:** Frontend (static S3/CloudFront) or simple client + backend API (Lambda + API Gateway) with at least 3 endpoints (CRUD).

2. **Storage & DB:** S3 for objects, DynamoDB for metadata.

3. **Auth:** Cognito user pool integration.

4. **CI/CD:** Automated pipeline (CodePipeline/GitHub Actions) with build/test/deploy; use IaC (CDK or CloudFormation).

5. **Monitoring:** CloudWatch dashboards, X-Ray traces, and at least one alarm & PagerDuty/Slack integration (or mocked webhook).

6. **Security:** IAM least-privilege roles, Secrets Manager/KMS usage.

7. **ML feature (optional but recommended):** Use Rekognition to tag images OR call a SageMaker endpoint for classification OR Lex chatbot for querying images. (This covers your Week 10 requirement.)
8. **Documentation:** README, architecture diagram, cost estimate, runbook, deployment steps, test plan, and a 10-minute recorded demo.

Team size: 2–3 students (or solo with higher expectations).
Assessment rubric (100 pts): Architecture & design 20 | Functionality 25 | CI/CD & IaC 15 | Security 15 | Observability & testing 10 | Presentation & docs 15.

Quick schedule & milestones (suggested)

- Week 0: Onboarding, create AWS accounts, share class repos.

- Week 4: Midterm check – students must have a working serverless API.

- Week 8: CI/CD & container milestone.

- Week 10: ML POC due (simple demo).

- Week 11: CV clinic & interview prep.

- Week 12: Final project presentations + final exam mocks.

Instructor resources & recommended study materials

- Official AWS exam guides (Cloud Practitioner & Developer Associate) — read fully and follow domain weights.
- AWS Skill Builder learning plans (Cloud Practitioner, Developer) and official practice question sets.